

Hydrodynamics and Acoustics of the Merging of a Drop with a Liquid in an Electrostatic Field

Yu. D. Chashechkin^{a,*} and V. E. Prokhorov^a

^a*Ishlinsky Institute for Problems in Mechanics of the Russian Academy of Sciences, Moscow, 119526 Russia*

**e-mail: yulidch@gmail.com*

Received November 1, 2025; revised November 10, 2025; accepted November 10, 2025

Abstract—The flow pattern and the sonic wave trains formed during the merging of a drop of distilled water with water surface in an electrostatic field with a potential of 2 kV synchronously recorded for the first time on the CDD (Charged Drop Dynamics) rig, which forms a part of the unique *GFK IPMech RAS* equipment. The experiments were carried out in the impact mode, when the falling drop forms a cavity, a splash, and a secondary cavity, from which a gas cavity separates and emits a retarding sonic wave packet. In an electrostatic field, a sound-emitting bubble of complex shape separates from the cylindrical end of the second cavity without forming a thin bridge observable when the bubble separates when the voltage is turned off. Selections from a video film illustrating the evolution of the fine flow structure during sound generation are presented together with the chronogram of the position of the lowest point of the cavity and an audiogram of the acoustic pressure. The primary and retarded sound packets, in which both the amplitudes and the basic frequencies vary nonmonotonically, are separated out and analyzed.

Keywords: drop, electrostatics, collision, cavity, bubble, sound packet

DOI: 10.1134/S0015462825603560

1. INTRODUCTION

Drop flows represent one of widespread forms of matter transfer in the atmosphere (individual drops, their groups, clouds, and cloud systems [1–3]) across interphase boundaries between the atmosphere, the hydrosphere, and the lithosphere. They can exist in the form of both falling rains and emitted sprays [4, 5] and in industrial large-scale technologies [6]; in recent years they are actively studied in the laboratory conditions [7]. The musical sounds of jet and drop flows reflected widely in the belletristics became the object of scientific investigations with the appearance of highly sensitive microphones and hydrophones in the mid-twentieth century [8]. The development of the sound recording techniques and the high-speed observations using first cinecameras and then videocameras [9, 10] opened the door for correlated investigation of hydrodynamics and acoustics of drop flows.

The audiograms of the sonic packets accompanying drop impacts, which are simultaneously recorded in both water and air media [11], include a short high-frequency primary package and retarded signals with different frequency compositions [12, 13]. The comparisons of the data of highly resolving video images of flow patterns and synchronized sound packets have shown that an important role in sound radiation is played by volume oscillations of gas cavities [14] formed in different stages of the evolution of a complicated multiphase flow picture [15].

The drop flows in the nature are often accompanied by charge separation and the appearance of regions with a high electric field strength, which can have a considerable effect on the drop dimensions and dynamics [16]. The atmospheric electricity shows itself in discharges of different types, both “quiet” glow and dangerous storms with thunder blasts [17]. The disintegration of a liquid jet flowing out of a nozzle into thin fibers and drops was established in investigations of the electrostatic field effect on fluid flows, which were started at the beginning of the last century [18]. The widely propagated scientific results favored the creation of certain large-scale technologies, among which there are the electrospinning, or mass production of fibrous materials [19], and the electrodispersion of solutions into fine droplets used in the mass-spectroscopic analysis [20]. The recent years demonstrated an active development of the theoretical and experimental electrohydrodynamics unifying the physics and mechanics of complicated fluids.

The field strength effect on the rate of drop separation from capillaries and their dimensions and shapes was traced in [21, 22].

The development of electrodynamic drop technologies requires the perfection of the methods of monitoring the state of a medium and the properties of the occurring processes, one of which is the recording of acoustic signals. This study deals with the experimental tracing of the electrostatic field effect on the properties of acoustic packages generated on the merging of an individual drop in the impact flow regime, when the kinetic energy of the drop is greater than its potential surface energy and the electrostatic component of the total energy.

2. PARAMETRIZATION

The performance of the experiments and an analysis of their results is based on the system of fundamental equations of heterogeneous fluid mechanics [23, 24] supplemented with the equation of state for the Gibbs potential G and its derivative, that is, the density ρ [25, 26], which is not presented here for the sake of brevity.

As follows from an analysis of the complete system of fluid mechanics equations, as applied to the problem of sound package generation in drop flows, the governing dimensional parameters include the Gibbs potentials of a drop G_d , the air medium G_a , and the target fluid G_t . The subscripts refer to the media characterized by the given parameter.

The governing parameters include also the derivatives of the Gibbs potential, namely, the density $\rho_{d,a,t}$, the pressure $P_{d,a,t}$, the temperature $T_{d,a,t}$, and other physical quantities, namely, the kinematic $\nu_{d,a,t}$ and dynamic $\mu_{d,a,t} = (\rho\nu)_{d,a,t}$ viscosities, the surface tension coefficients, both total σ_d^a , σ_t^a and normalized by the medium density $\gamma_d^a = \sigma_d^a/\rho_d$, $\gamma_t^a = \sigma_t^a/\rho_t$; the equivalent diameter D , the surface area S_D , the volume V_D , the mass $M = \rho V_D$, the momentum $p_D = MU$, and the vertical velocity U of the drop at the moment of the primary contact with the target fluid.

The basic parameters also include the extensive kinetic energy at the moment of the contact $En_k = MU^2/2$ and the potential surface energy (PSE) $En_\sigma = \sigma_d^a S_D$ concentrated in a thin surface layer, whose thickness is of the order of the molecular cluster dimension, $\delta_\sigma \sim 10^{-6}$ cm. The PSE is characterized by its high density $W_D^\sigma = En_\sigma/V_\sigma$, as distinct from the kinetic energy distributed throughout the entire drop bulk with the density $W_D^k = En_k/V_D$. The ratio of the drop energy components $R_\sigma^k = En_k^k/En_D^\sigma$ can be both large and small but the ratio of the densities of the energy components $R_W = W_D^k/W_D^\sigma \sim \delta_\sigma/D$ is always a small quantity under the experimental conditions of this study.

The experimental investigation of the electrostatic field effect on the structure and dynamics of drop flows, which was started at the beginning of the last century [18], is being continued basing on a new technological foundation [27]. In describing the experimental conditions the baseline dimensional parameters

include the electrostatic field energy $En_e = \frac{\epsilon\epsilon_0 V_D}{2} \left(\frac{\Phi_e}{H_e} \right)^2$, whose value is determined by the drop volume V_D and the strength of the applied field $E_e = \Phi_e/H_e$, which is assumed to be homogeneous. Here, ϵ is the relative electric permeability of the drop fluid, ϵ_0 is the universal dielectric constant, and H_e is the distance between the electrodes under the voltage Φ_e .

The set of the energy parameters includes also the potential energy $En_p = MgD$ in the gravity field with the gravity acceleration g on a scale D .

The ratios of the characteristic parameters preassign the set of dimensionless parameters, which includes conventionally the numbers of Reynolds $Re = UD/\nu$ based on the contact velocity of the drop U , Froude $Fr = E_k/E_p = U^2/gD$, Bond $Bo = gD^2/\gamma$, Ohnesorge $Oh = \nu/\sqrt{\gamma D}$, and Weber, both capillary

$We = DU^2/\gamma$ and electric $We_e = \frac{\epsilon\epsilon_0}{\rho U^2} \left(\frac{\Phi_e}{H_e} \right)^2$, which are used for the comparative analysis of the results of independently performed experiments.

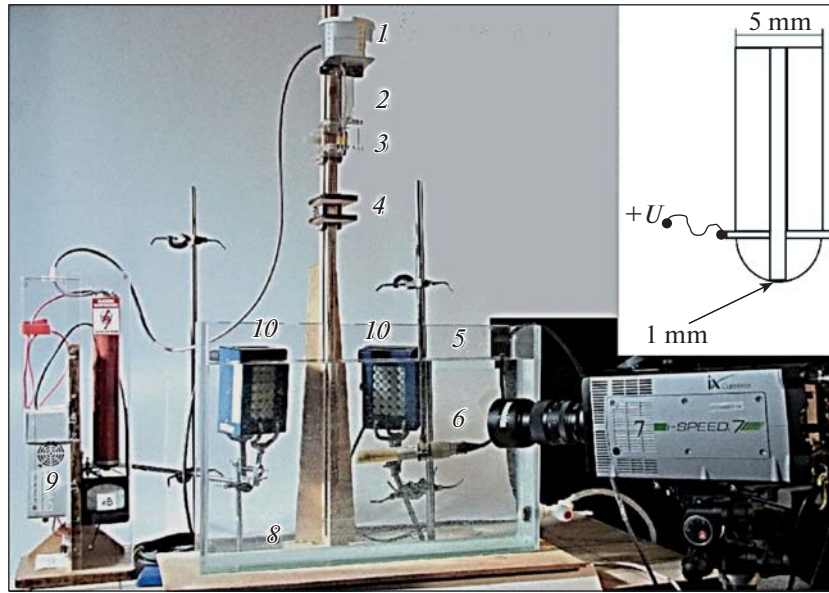


Fig. 1. Photo of the experimental setup.

The additional dimensionless parameters are also the ratios of the drop energy components, that is, the kinetic and potential surface energies $R_{En}^{k\sigma} = En_k / En_\sigma$ (for a spherical drop $R_{En}^{k\sigma} = DU^2 / 12\gamma$), the kinetic and electric energies $R_{En}^{ke} = En_k / En_e$, and the potential surface energy and the electric energy $R_{En}^{ke} = En_\sigma / En_e$.

The large number of dimensionless parameters reflects the complexity of the occurring processes, which form a rapidly evolving three-dimensional flow pattern and the variety of their spatial and temporal scales. The high densities of the components of the potential surface and electric energies ensure the possible violation of surface continuity and fluid disintegration into individual jetlets observable in the pictures of the confluence of fluid drops in the impact regime [28]. In these experiments, we studied for the first time the electric field effect on both the flow pattern occurring on the confluence of a drop in fall and the acoustic signal properties.

3. EXPERIMENTAL SETUP

The experiments were performed on the modified CDD (Charged Drop Dynamics) rig, which is the part of *GFK IPMech RAS* [29]. The setup, whose photo is presented in Fig. 1, includes the hydraulic, electric, acoustic, and optical blocks and a block of data gathering and preparing

The hydraulic block includes conventionally a storage 1, the fluid admission system with the flow rate control 2, and a dripper 3 with exchangeable capillary tubes with an electrode (anode) connected with the positive pole of the source of high constant voltage. The drop separated from the capillary tube under its own weight falls freely and interrupts on its way the light beam in the flight recording device 4, whose photoelectric receiver forms a pulse triggering the videofilming and recording of acoustic signals. The drop falls into a transparent basin 5 filled by tap water and measuring $60 \times 40 \times 10 \text{ cm}^3$, where a hydrophone 6 is placed. The picture of the flow produced in the basin by the downfallen drop is recorded by the video-camera 7.

The electrode 8, which is a cathode lying on the basin 5 bottom, is connected with the negative pole of the controlled source of high constant voltage 9. An anode made in the form of hemispherical tip of dripper 3 is connected with the positive pole of the source (see insert in Fig. 1). The range of the voltage controlling the source $\Phi_e = 0\text{--}30 \text{ kV}$.

The acoustic block includes a GI-54 hydrophone (the bandpass is 0.002–100 kHz and the nonuniformity is 3 dB), whose outlet is connected to an 8-channel data preparation block (DPB) synchronizing the processes of recording the video and acoustic signals. The block converts the analog signal of the hydro-

phone into the binary 12-digit code (the discretization frequency is up to 58 MHz), which is transmitted into a computer, together with the sensitive information.

The optical unit includes lighters 10 (of the Multiled type) and the i-Speed 717 videocamera 7 (iX-Cameras), which is controlled by the home-made data preparation block (DPB). The videostream arrives into a computer, where it is converted into the standard format and is conserved in a storage device. In certain experiments the flow picture was additionally illuminated by photodiode Godox projectors placed behind an opaque screen.

In the experiments the water started to arrive into the dripper 3 after high voltage has been turned on. A drop of distilled water separated from the capillary tube of the dripper crossed the light beam and fell into the basin filled by tap water. The signal from the photoelectric receiver started the videocamera and the hydrophone. The error of synchronization of the acoustic and optical signals amounted to 20 μ s. The experiments were carried out at the difference of potentials $\Phi_e = 2$ kV. The videofilming frequency, when recording the flow pictures was 50000 frames per second and the resolution was 380×456 (the absolute resolution was 0.06 mm). The results of the investigation of the electrostatic field effect on the drop separation were presented in [22].

4. FLOW PICTURES AND AUDIOGRAM OF ACOUSTIC SIGNALS IN AN ELECTROSTATIC FIELD WITH THE POTENTIAL $\Phi_e = 2$ kV

The evolution of the flow picture occurring in a target fluid on the merging of a drop, $D = 5.0$ mm in diameter, falling at the contact velocity $U = 2.4$ m/s, in an electric field with the potential $\Phi_e = 2$ kV is illustrated in the selection from the videogram presented in Fig. 2. The time dependence of the position of the lower terminal point of the deformed fluid surface is presented in Fig. 2i.

The drop merging in the impact regime deforms the free surface of the target fluid and forms a cavity [28]. In the stage of the greatest depression the primary cavity, whose image is presented in Fig. 2a, is crossed by a system of slightly expressed horizontal bands, which visualize a group of capillary waves shed from the crown edge. The wave length ($\lambda = 1.4$ mm at $t = 23.2$ ms, Fig. 2a) amounts to $\lambda = 2$ mm at $t = 43.2$ ms. The cavity, which reaches its maximum depth $h_1 = 11.2$ mm at $t = 23.2$ ms (label 2 in Fig. 2i), collapses then to the depth $h_1 = 3.0$ mm ($t = 55.2$ ms, time is measured from the moment of the primary contact of the drop shown by label 3 on the chronogram in Fig. 2i) and goes over into a slowly evolving depression, whose depth varies irregularly near the approximating dependence $h_1 = -0.016t + 4.22$ mm on the $55.2 < t < 137.2$ ms interval (labels 3 and 4).

A splash of conical shape is formed on the fluid surface in the process of cavity collapsing ($t = 142.2$ ms, Fig. 2b) and a comparatively large drop is thrown from its vertex.

Then a splash, which starts to subside, forces under its own weight the slowly raising depression bottom and forms an annular cavity, which embraces its base. In the technique realized in this experiment only the external contour of the cavity is visualized in the side-drawn image in the transmitted light (Fig. 2c). The depression depth amounts to the local value $h_t = 4.8$ mm at $t = 153$ ms (label 5 in Fig. 2i) and starts to decrease, initially rapidly and then more slowly to the moment $t = 160.8$ ms (label 6), when a new cavity appears in the field of sight; it is generated by the descending cylindrical body of the splash.

At the same time a returning secondary drop of ovoidal shape and $D_2 = 6.6$ mm in diameter, appears in the flow picture; it was thrown out earlier from the splash vertex. It is accompanied by a second cylindrical cavity formed by the submerging cylindrical portion of the splash. The plane bottom of the second developing cavity is largely coated by small-size gas bubbles. The second cavity generated by the declining splash starts to be noticeable on the chronogram at $t = 160.8$ ms as a sharp increase of the free surface disturbance depth, whose maximum value $h_2 = 8.3$ mm is achieved at $t = 172.4$ ms (label 7).

A videoframe of the flow picture is presented in Fig. 2e.

The flows generated in the target fluid by the merging drop and the action of electric fields cause the separation of the lower extended cylindrical region of the cavity and the formation of a gas cavity (the separation process is illustrated in more detail in Fig. 3). As this passes, the cavity depth decreases sharply and amounts to the value $h_2 = 1.3$ mm at $t = 179.8$ ms.

The separated portion of the cavity forms a gas bubble of complicated shape with a roundish bottom and a plane vertex, whose shape changes rapidly and irregularly. The bubble, whose vertex and side surfaces are adjoined by gas bubbles, is shown in Fig. 2f. The second cavity collapses initially fairly rapidly

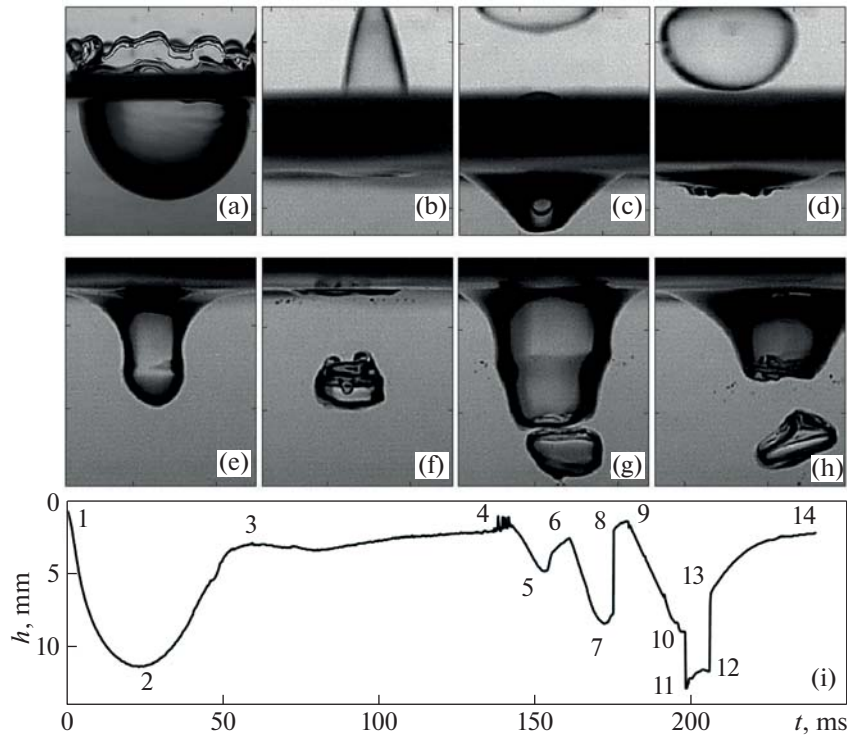


Fig. 2. Evolution of the flow pattern on the confluence of a fluid drop falling in an electrostatic field with the potential $\Phi_e = 2$ kV in the impact confluence regime: (a–h) selections from the videogram of the flow picture at $t = 23.2, 142.2, 153.0, 161.0, 172.4, 179.8, 198.4$, and 205.8 ms; (i) time dependence of the depth of the lower edge of the deformed fluid surface ($D = 5.0$ mm, $U = 2.4$ m/s, $Re = 11950$, $Fr = 117$, $Bo = 3.3$, $Oh = 1.6 \times 10^{-3}$, $We = 390$, $We_e = 2.0 \times 10^{-6}$, $R_{En}^{k\sigma} = 33$, and $R_{En}^{ke} = 5 \times 10^5$; the scales of the images are individual and the scale division is 5 mm).

(during $\Delta t = 7.4$ ms) and then more slowly until the horizon of the emerging depression is reached at the depth $h_2 = 1.9$ mm (label 8, $t = 175.2$ ms).

After the splash immersion a remainder of the depression converging to the fluid surface is conserved in the fluid; it is approached by the drop thrown earlier from the splash vertex. The drop submerging into the remainder of the second cavity touches its side wall and forms a third cavity. The lateral surface of the drop, which is in contact with the cavity wall, forms a “confluence band” of large area. The converted PSE portion of the eliminated surfaces and the transmitted kinetic energy of the drop portion take part in the formation of the rapidly growing third cavity, which intrudes actively into the fluid thickness (Fig. 2g). The cylindrical cavity depth increases sharply, so that $h_3 = 8.9$ mm at $t = 197.8$ ms (label 10 in Fig. 2i). This is followed by a complicated process of the contact between the cavity bottom and the surface of the oscillating gas cavity with the formation of the common gas cavity.

In the contact stage the bottom of the growing cavity approaches the upper region of the gas bubble, whose position is supported by the flows, which have been formed. As this passes, the close-together surfaces are synchronously curved. They perform complicated motions, in the process of which they touch locally each other, then separate, and contact with each other again forming fine, rapidly evolving structures. At $t = 198.4$ ms the cavity and the gas bubble are definitely joined forming an extended cavity with the depth $h_3 = 12.8$ mm (label 11). After the bubble detachment at $t = 205.8$ ms (label 12) the cavity depth reduces sharply to $h_3 = 11.6$ mm and then reaches slowly the level of the depression bottom, which continues to float up ($t = 206.6$, label 13).

The chronogram presented in Fig. 2i illustrates the general evolution of the position of the greatest depression on the free surface. The key flow element in the initial stage is the cavity (Fig. 2a), which reaches its maximum depth, $h_1 = 11.3$ mm, at $t = 23.2$ ms (label 2 in Fig. 2i). Further on the cavity starts to descend on the free surface, whereupon a depression region is simultaneously formed behind the tail of

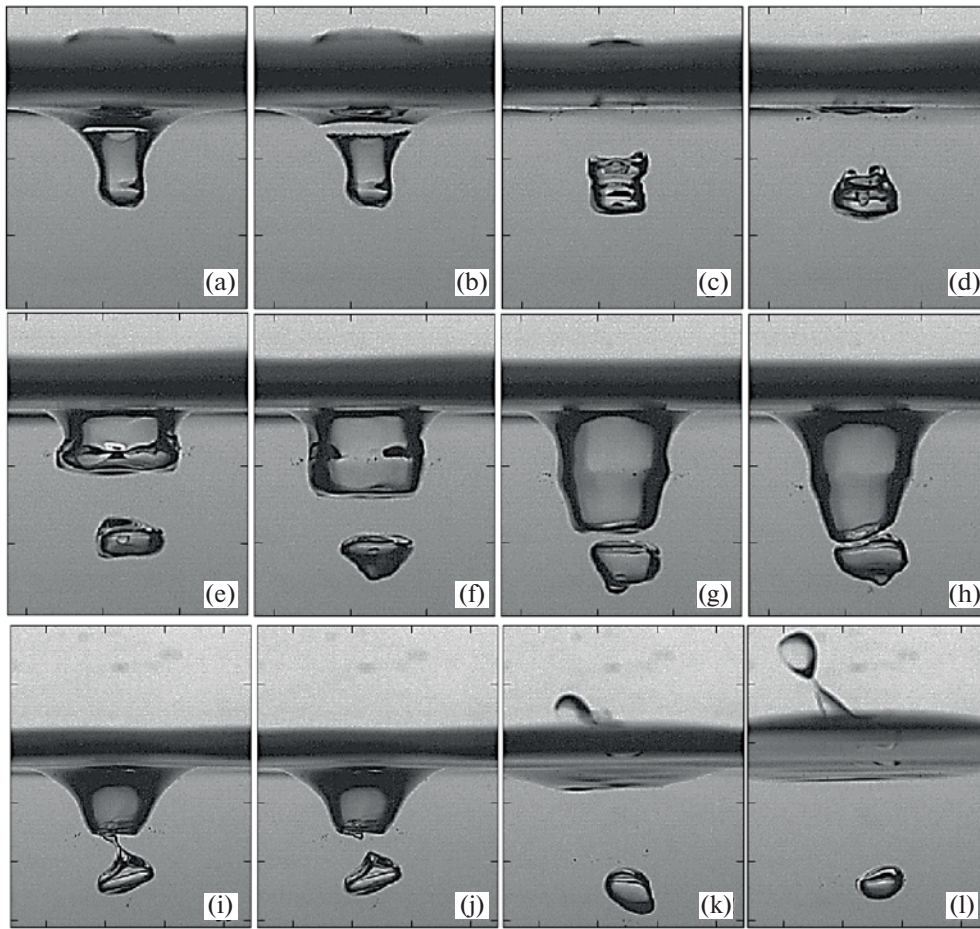


Fig. 3. Evolution of the flow pattern in the process of separation and reattachment of a sound-emitting gas cavity on the confluence of a drop falling in an electrostatic field with the potential $\Phi_e = 2$ kV; (a–l) $t = 175.2, 175.4, 178.5, 180.8, 188.4, 192.1, 197.3, 201, 205.9, 206.2, 224.5,$ and 243 ms; the scale division is 5 mm.

a group of diverging annular capillary waves. Starting from $t = 55.2$ ms (label 3) the position of the maximum depth point is governed by the dynamics of the slowly raising depression, whose plane bottom is slowly displaced toward the free surface.

The behavior of the curve changes sharply at $t = 142.6$ ms, when the bottom of an annular depression appears in the field of sight; it embraces the submerging pedestal and the splash base (label 4, $t = 137.8$ ms). The maximum value of the depression depth, $h_1 = 4.7$ mm, is reached at $t = 153.4$ ms (Fig. 2c and label 5 in Fig. 2i). Then the depression is rapidly filled up to the moment $t = 160.6$ ms (label 6 in Fig. 2i), when a rapidly deepening second cavity initiated by the immersion of the cylindrical portion of the splash appears in the field of sight (Fig. 2f). After a maximum depth has been achieved ($h = 8.4$ mm, $t = 172.2$ ms) the second cavity starts to collapse intensely up to the moment $t = 175.8$ ms ($h_2 = 1.8$ mm), when it reaches the level of the ascending depression bottom.

After a short time interval, at $t = 179.8$ ms (label 9) the sharply changing nature of the chronogram reflects the appearance of a third cavity, whose depth amounts to $h_3 = 8.9$ mm ($t = 178.0$ ms). At this moment the cavity bottom touches the upper edge of the floating-up gas bubble. The common depth of the deformed contact surface starts to increase sharply (label 10) and amounts to the value $h_3 = 12.8$ mm at $t = 198.6$ ms (label 11). After a short interval of the common existence of the cavity and the bubble the joint action of the field and local flows disintegrates the design and separates the gas bubble (Fig. 2h, label 12, $t = 205.4$ ms). The depth of the truncated cavity decreases sharply at $t = 206.4$ ms (label 13),

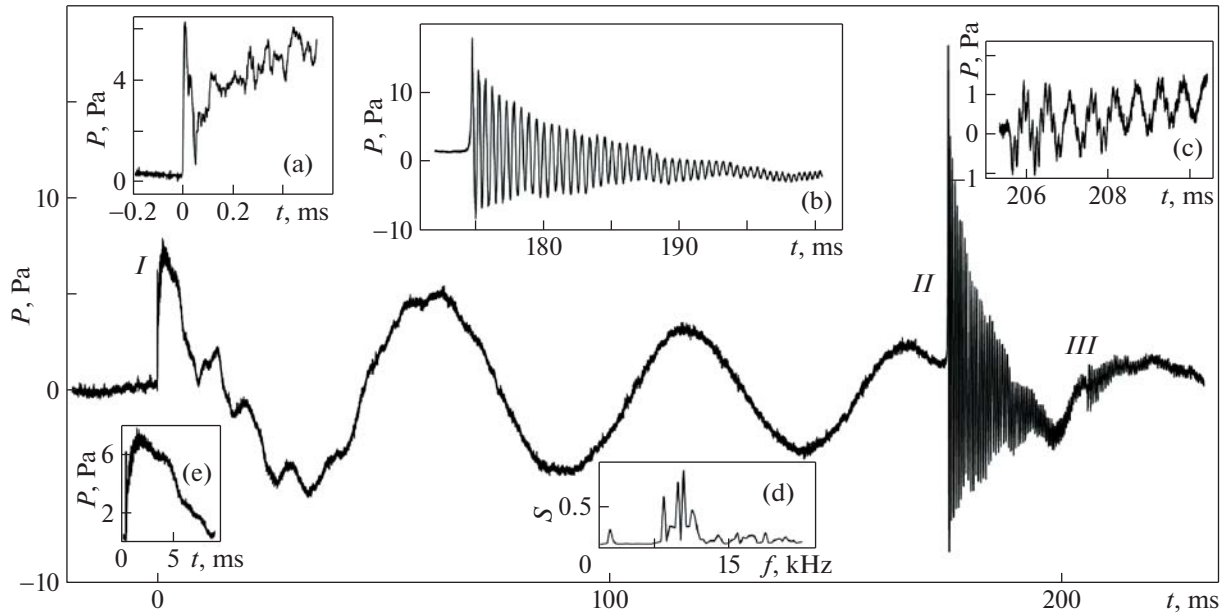


Fig. 4. Audiogram (a) and selections of the primary pulse (*I*), the resonance package (*II*), and its high-frequency addition (*III*) illustrating the flow patterns for $\Phi_e = 2$ kV; (b–e) relate to $t = 193.4, 193.8, 205.4$, and 224.2 ms; the scale division is 2 mm.

whereupon its bottom reaches smoothly the level of the bottom of the floating-up depression. The cavity depth amounts to $h_3 = 2.1$ mm at $t = 240$ ms (label 13 in Fig. 4i).

Generally, the depression bottom geometry interpolated on the time interval $t = 58.6$ – 240 ms (without account for the contribution of the cavities) can be approximated by the line $h_e = a_e t + b_e$ ($a_e = -0.004$ mm/ms, $b_e = 3.0$ mm), whose inclination is considerably smaller than that on the drop confluence in the absence of the field.

The earlier-performed investigations showed that on the contact of a solitary drop with a fluid at rest in the absence of an external electrostatic field the acoustic signal includes two types of successive sound packets, namely, a high-frequency signal appearing at the moment of the primary contact between the drop and the target fluid surface and a comparatively low-frequency (resonant) signal radiated with a delay determined by the duration of the formation process of the sound-radiating bubble [11–15].

As follows from an analysis of the audiogram reflecting the temporal variability of the acoustic pressure, the general structure of the signal is conserved on the confluence of the drop in fall in the electrostatic field with the potential $\Phi_e = 2$ kV. However, the parameters of the acoustic packets vary, since the field has a considerable effect on the free surface displacement and geometry. The distinctive features of the separation process of a gas cavity radiating a sound packet in an electrostatic field are illustrated by the selection from a videofilm presented in Fig. 4.

In the absence of an electrostatic field the separation of a radiating gas bubble is preceded by the deformation of the second cavity, whose bottom extends and takes the cylindrical shape. Then a small band appears on the cylinder surface; its diameter decreases progressively. The regions of the cylindrical cavity adjoining the band take the conical shape. They remain to be connected by a thin channel, or neck [14]. The sound radiation starts after the neck disruption.

The electrostatic field, even relatively weak, with the potential $\Phi_e = 2$ kV changes considerably the picture of the gas cavity separation presented in Fig. 2e. In an electrostatic field the lower, cylindrical portion of the second cavity with the rounded base separates as integral whole without the formation of a constricting neck. The cavity breaks up at only one time throughout the entire section without changing the shape of its external contour (Fig. 3a). The radiation of the resonance sound packet *II* (Fig. 4), whose audiogram is presented in Fig. 4, starts at the moment of the disruption of the cylindrical region of the cavity and the detachment of its lower region. In these experiments the time interval between the disruption on the cavity body to the complete bubble separation amounts to 140 μ s.

The diverging surfaces of the cavity bottom and the vertex of the separated gas cavity take concave shapes (Fig. 3b). A group of tiny gas bubbles is conserved on the sharp edge of the bubble vertex. The separated gas cavity with “glued” tiny bubbles submerges slowly under the action of background flows. The tip of a submerging secondary drop, thrown earlier from the drop vertex can be traced above the free surface (Fig. 3b). The remainder of the second cavity descends on the free surface.

The shape of the separated gas cavity with capillary waves running over its surface varies continuously. The bubble almost stops and its shape becomes near-cylindrical, with a roundish bottom and a flat vertex (Fig. 3c). The phase surfaces of the waves visualize the alternating dark and light strips. A few gas bubbles, larger than those in Fig. 3b, are conserved on the upper edge of the cavity. The cavity bottom oscillates strongly: convex in Figs. 3a and 3b, it becomes concave at its center, as in Fig. 3c. The positions and dimensions of individual gas bubbles change continuously.

The cavity becomes gradually lens-shaped; its bottom becomes convex but its vertex at the center is concave with increasingly larger gas bubbles on its edge (Fig. 3d). The tiny bubbles move along the edge merging continuously into progressively larger bubbles. The submerging secondary drop forms a new cavity, whose base can be seen in Fig. 3d.

The drifting gas bubble takes gradually an elliptic shape distorted by individual small-scale disturbances (convexity in the left upper corner and a group of capillary waves in the right upper corner). Individual gas bubbles move over its surface (at the center and at the top on the left). The growing cavity takes a complicated shape: its upper region is cylindrical, an annular rim can be traced in the lower region, and a conical depression is at the center of the bottom. The cavity edge is deformed by capillary waves (Fig. 3e).

The shape of the growing cavity becomes gradually smoother and the rim extends leaving a small thickening. The gas cavity under the cavity bottom takes the shape of an overturned cone with gas bubbles being displaced and merged over its upper edge and walls (Fig. 3f).

Then the lower portion of the growing cavity takes the conical shape with a plane base. The rapidly growing cavity approaches the gas bubble, whose shape has been changed, whereas its spatial position has remained almost unchanged. Several merging gas bubbles form a characteristic convexity on the bubble bottom, which separates with time and forms a new tiny bubble of spheroidal shape (Fig. 3g).

As the third cavity approaches the bubble, a complicated synchronized variation of the shapes of the closing together surfaces can be observable; being still separated they become in turn convex and concave (Fig. 3h). As this passes, an active motion of tiny bubbles can be observable on the bubble surface. The duration of the interval of the interaction between the closing-together surfaces $\Delta t_i = 9$ ms.

During these intense displacements the gas cavity surface extends and touches the cavity bottom (Fig. 2g), which starts to descend on the free surface. A thin gas channel connecting the vertices of the conical tips of both cavities (dam) becomes thinner in its central region (Fig. 3i) and breaks (Fig. 3j).

The conical appendices shrink gradually into a collapsing cavity and a drifting gas bubble. The breaking of a thin air channel connecting the conical cusps of a portion of the cavity bottom and the separating gas cavity is accompanied by the excitation of bubble oscillations and the generation of a new sound package *III* (Fig. 4).

The separated bubble evolves slowly; short capillary waves run over its surface. The plane bottom of the cavity bulges gradually upward and forms a growing streamer (Fig. 3k). Then a drop is thrown from the streamer vertex; initially it is connected with its body by a thin bridge. The drifting gas cavity becomes smoother, while capillary waves continue to run over its surface.

The audiogram of the acoustic signal recorded by the hydrophone on the merging of a falling drop of distilled water with the tap water in a basin in the case of turned-on electric field at the difference of potentials $\Phi_e = 2$ kV, which is presented in Fig. 4, retains the key features of the signal noted in recording sound packages in the absence of the field [11–15]. A short high-frequency sound package *I* begins to be radiated at the moment of the primary contact of the drop with the target fluid. A lower-frequency (resonance) package *II* is formed, when the second gas cavity breaks up. After the decay of signal *II*, when the bubble attached to the bottom of the second, rapidly collapsing cavity separates from it, package *III* is generated.

The primary sound package, whose developments are presented in inserts (a, e), includes the initial pulse of amplitude $P = 6.2$ Pa, with the leading front duration $\tau_f = 7$ μ s (insert (a)). It is adjoined by a more prolonged package with the amplitude $P = 6.8$ Pa and the leading front duration $\tau_{fp} = 1$ ms (insert (e)). The frequencies $f_i = 1.3, 7.5, 9.1,$ and 9.8 kHz are highlighted in the oscillation spectrum on the $0.15 < \Delta t < 3.2$ ms interval (insert (d)).

The primary package changes for a group of low-frequency oscillations of hydrodynamic origin; these are due to the formation of disturbances of the free surface shape and location. Their maxima proceed with

the intervals $T_1 = 60, 56, 50$, and 53 ms. Additional disturbances appear with the drop of the first oscillation; their maximum values follow at the shorter intervals $T_2 = 11, 6, 10$, and 10 ms (the first maximum is the same for both groups). The oscillations are associated with the formation of a group of surface disturbances, which includes a cavity, a crown, a splash, diverging annular capillary waves, and a depression.

The resonant package *II* with a maximum amplitude $P_s = 26$ Pa is recorded with a delay $\Delta t = 174.5$ ms, when the primary cavity, the growing and decaying splashes, the annular depression, and the second cavity are formed. The signal formation is in agreement with the appearance of a discontinuity of the primary cavity (Fig. 3b). As distinct from the flows with turned-off field [3, 14], the detachment of the gas cavity from the cavity base occurs without formation of a thinned region (neck). The joint action of the flows and the electrostatic field results in the “cutting” of the cavity and the formation of a separating gas cavity with sharp edges and a flat vertex.

An analysis of the acoustic signal, which is presented in Fig. 4 in the form of an extended plot of sound pressure variations, given in Fig. 4 (insert (b)) shows that its frequency f_{II} increases smoothly from 2030 Hz at the beginning to 2100 Hz at the end of the package. The $\Delta f_{II} = 70$ Hz variation can be due to the smoothing of the shape and a decrease of the radiating gas cavity volume.

The signal amplitude decays uniformly. An initial region of rapid signal decay is expressed in the development view of signal *II*; further on the decrease rate is reduced sharply. The acoustic radiation almost ceases, when the bubble starts to touch the incident cavity.

Tiny gas bubbles can be traced on large magnification on the contour of the plane bubble vertex. The cavity bottom raises rapidly, while the bubble submerges with the flow conserving its flat vertex and sharpened edges (Fig. 3c).

The flow pattern changes rapidly on the $197.3 < t < 224.2$ ms time interval. The rapidly growing cavity with the flat bottom (Fig. 3g) approaches the gas cavity of complicated shape, whose position is fixed by the developed flows. The upper protrusion on the bubble approaches the cavity bottom and touches it over a small surface (Fig. 3h). The arrangement of the phase surfaces between the cavity and the bubble changes rapidly and irregularly. Then the cavity, whose bottom takes partially the conical shape, starts to descend on the free surface and leaves a sharp-edged protrusion tied with a conical protrusion on the bubble by a thin waist.

The neck disruption is accompanied by the radiation of a new resonant sound package *III*, which appears with the delay $\Delta t = 31$ ms ($t = 205.5$ ms). The shape of the separated bubble changes (Figs. 3i and 3j).

The structure of the extended signal in this figure (insert *c*) indicates the complexity and multiple-frequency capability of the source of radiation of the sound package *III*, which can produce oscillations of several bound gas cavities, whose shapes vary rapidly, with capillary waves running over their surfaces (Figs. 3i and 3j). The signal *III* frequency in the initial region $f_{III} = 1860$ Hz, that in the mid region is 1630 Hz, and that in the tail region is 1820 Hz. The signal spectrum contains also the high-frequency oscillations with $f_{III} = 9400$ Hz, whose existence can be attributed to the radiation of one of tiny bubbles covering the uneven surface of the contracting cavity bottom (Fig. 4h). All acoustic and hydrodynamic disturbances decay with time.

SUMMARY

The CDD (Charged Drop Dynamics) rig, which forms a part of the unique *GFK IPMech RAS* equipment, allows to study consistently the structure of the flow formed in the target fluid by a free-falling drop in an electrostatic field and to record the accompanying acoustic signals.

The drop merging picture in the impact regime in an electrostatic field with the voltage of 2 kV is for the first time consistently recorded and a considerable field effect on separation pattern of the radiating gas cavity is established. In the absence of the field the cavity separates with the formation of a thinned neck, whereas in an electric field it detaches without changing the general contour of the cavity.

The primary acoustic signal in an electric field is supplemented by two resonant acoustic packages, whose frequencies vary with time, which points at time-dependent variations of the radiating gas cavity volume in the electrostatic field.

ACKNOWLEDGEMENTS

The authors wish to thank N.G. Soloviev for assistance in organizing the experiments

FUNDING

The experiments were performed on the CDD (Charged Drop Dynamics) rig, which forms a part of the unique *GFK IPMech RAS* equipment, within the framework of the state-financed theme “Development of analytical and numerical algorithms for studying the processes in continuum mechanics, thermodynamics, and quantum mechanics, investigating wave, convection, thermal, and fragmentation processes in complicated fluids, and developing conjugate methods of theoretical and experimental studies of mechanics and thermodynamics of fluid flows” (State Registry no. 124012500442-3).

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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Translated by M. Lebedev

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